HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and deployed **AI-powered agents** and intelligent workflows using **Python**, integrating with **LLMs** to automate business processes, reducing manual intervention by an estimated **22%**.
- Contributed to the development of **Retrieval-Augmented Generation (RAG)** solutions, enabling efficient querying of large document sets and improving information retrieval accuracy for internal knowledge bases.
- Designed and implemented **REST APIs** in **Java Spring Boot** for seamless integration between enterprise applications and new AI services, ensuring robust data exchange and service reliability.
- Participated in **CI/CD** pipeline development and maintenance using **Jenkins** and **Git**, facilitating frequent and reliable code deployments to cloud environments.
- Collaborated with architects and product managers to define technical requirements for AI-native applications, translating business needs into actionable development tasks.
- Applied **responsible AI** principles and security best practices throughout the software development lifecycle, ensuring ethical and secure AI solution deployment.
- Containerized microservices using **Docker** and deployed them to a scalable **AWS** environment, managing configurations and ensuring operational readiness.

SKILLS

AI/ML Frameworks – Generative AI, LLMs, RAG, AI Agents, Copilots, Prompt Engineering, LangChain, LlamaIndex, Semantic Kernel, PyTorch, TensorFlow, scikit-learn.

Programming Languages – Python, Java, JavaScript, TypeScript, C++.

Backend & Cloud – Spring Boot, REST APIs, Microservices, AWS, Azure, GCP, Docker, Kubernetes, Event-Driven Architecture.

Data & Databases – SQL, PostgreSQL, MySQL, Oracle, Data Warehousing, Data Engineering.

Development Practices – CI/CD, Jenkins, Git, Agile, Testing, System Design, Observability.

PROJECTS

AI-Powered Customer Support Assistant (HCL Technologies)

- Developed a prototype **AI assistant** using **Python** and **LangChain** to handle common customer inquiries by integrating with an internal knowledge base via **RAG**.
- Engineered a **FastAPI** interface to serve the AI assistant, processing an average of **150 requests per hour** during the pilot phase.
- Implemented prompt engineering techniques to refine user query understanding and improve the relevance of generated responses, reducing follow-up queries by **18%**.

Intelligent Document Processing Pipeline (MSc Applied Project)

- Built a **Python** pipeline to extract and classify information from enterprise documents using **NLP** techniques and a fine-tuned transformer model.
- Applied **scikit-learn** for text vectorization and classification, achieving an accuracy of **89%** on a dataset of 5,000 documents.
- Designed a **Power BI** dashboard to visualize extracted data and process completion status for business stakeholders.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)